Reliability and Cronbach's Alpha

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Introduction

Cronbach's alpha, introduced by Lee Cronbach in 1951, summarises the internal consistency of a multi-item scale by quantifying how strongly the items co-vary after accounting for the total number of items. The intuition is that items intended to measure the same underlying construct should correlate with each other; alpha rises with the average inter-item correlation and with the number of items. Cronbach's alpha is now ubiquitous in questionnaire validation, patient-reported-outcome (PRO) instrument development, psychometric evaluation of clinical scales, and any setting where a composite score is formed from multiple ordinal or continuous items. Despite well-known statistical limitations, it remains the single most reported reliability statistic in the clinical-research literature.

Prerequisites

A working understanding of classical test theory, the concept of a true score plus measurement error, and the construction of composite scores from multi-item rating scales.

Theory

Cronbach's alpha is

$$\alpha = \frac{k}{k-1}\left(1 - \frac{\sum_{i=1}^{k} \sigma_i^2}{\sigma_T^2}\right),$$

with k the number of items, σ_i^2 the variance of item i, and σ_T^2 the variance of the total summed score. The statistic ranges from $-\infty$ (theoretically; in practice 0) to 1. Conventional thresholds are 0.70 (acceptable), 0.80 (good), and 0.90 (excellent), with values above 0.95 typically indicating redundant items rather than superior reliability. McDonald's omega is a more flexible alternative when the assumption of tau-equivalence (equal true-score variances across items) is doubtful.

Assumptions

Items are tau-equivalent (each item measures the same true construct with the same loading), the scale is unidimensional (a single underlying factor explains the inter-item covariance structure), and items are continuous or quasi-continuous (Likert with at least 5 levels). Violations are common, and McDonald's omega or hierarchical-omega estimators give a more honest reliability estimate when the assumptions are not met.

R Implementation

```r
library(psych)

set.seed(2026)
n <- 200
theta <- rnorm(n)
items <- sapply(1:8, function(i) 0.7 * theta + rnorm(n, 0, 0.7))
```

```r
colnames(items) <- paste0("q", 1:8)

alpha_res <- psych::alpha(items)
alpha_res$total
alpha_res$alpha.drop[, c("raw_alpha", "std.alpha")]
```

Output & Results

`psych::alpha()` returns the raw and standardised Cronbach's alpha for the full scale, plus an "alpha drop" table showing how alpha would change if each item were removed. A large positive drop (alpha increases without an item) flags that item as inconsistent with the rest of the scale and a candidate for revision or removal. The output also includes 95 % confidence intervals (via Feldt's method) and the average inter-item correlation, which is often more informative than alpha itself.

Interpretation

A reporting sentence: "The eight-item PRO scale had Cronbach's alpha 0.82 (95 % CI 0.78 to 0.86, Feldt method), indicating good internal consistency. The average inter-item correlation was 0.36, supporting the tau-equivalence assumption qualitatively. No single item substantially changed alpha when dropped (largest drop -0.01), so all items were retained for the final scale. McDonald's omega-total was 0.83, in close agreement with alpha." Always report both alpha and omega when feasible.

Practical Tips

- Alpha depends on the number of items: long scales inflate alpha mechanically, even when items are only weakly inter-correlated. Reporting the average inter-item correlation alongside alpha gives readers a fairer picture of true item coherence.
- Low alpha (< 0.70) may reflect a multi-dimensional scale rather than unreliable items. Always check the factor structure with exploratory or confirmatory factor analysis before concluding that items are unreliable.
- Very high alpha (> 0.95) suggests redundant items measuring essentially the same content; consider trimming the scale by removing the most redundant items, which improves administrative efficiency without sacrificing reliability.
- For ordinal items with fewer than five categories, the standard Pearson-correlation-based alpha is biased downward; use ordinal alpha (computed from polychoric correlations) or McDonald's omega instead.
- Pair Cronbach's alpha with confirmatory factor analysis (CFA) to verify the assumed unidimensional structure; alpha computed on a multidimensional scale is uninterpretable as a reliability statistic.
- For test-retest reliability and inter-rater reliability, the appropriate statistics are the intraclass correlation coefficient (ICC) and Cohen's or Fleiss's kappa, respectively; alpha measures only internal consistency, not stability or agreement.

R Packages Used

`psych::alpha()` for the canonical Cronbach's alpha with drop analysis, `psych::omega()` for McDonald's omega and hierarchical-omega; `ltm::cronbach.alpha()` as a lightweight alternative; `MBESS::ci.reliability()` for advanced CIs (Feldt, bootstrap, Bonett); `lavaan` for confirmatory factor analysis to verify the unidimensional assumption.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale